

Lesson 1: Introduction to Data Analysis with Python and Pandas

Student Study Material for Moodle Upload

Purpose: This introductory lesson explains what data analysis is, where datasets come from, how data can be arranged, and how Pandas can be used to load and inspect a small dataset.

Learning outcomes

- Explain what data analysis means.
- Distinguish between data, information, and insight.
- Identify rows, columns, variables, observations, values, and missing values.
- Describe common dataset sources and basic dataset quality questions.
- Explain the difference between wide data and long data.
- Describe the basic data analysis workflow and explain what happens during each phase.
- Explain why Python and Pandas are useful for introductory data analysis.
- Load a simple CSV file with Pandas and inspect it using basic commands.

Important idea

Data analysis is not only about writing code. It is about using data to answer questions and then explaining what the results mean.

1. What is data analysis?

Data analysis is the process of using data to answer questions, solve problems, and support decisions. A table full of numbers, names, dates, or categories may not immediately tell us anything useful. Data analysis helps us turn raw data into useful information and then into insight.

For example, a business may have a list of sales transactions. Through data analysis, the business can answer questions such as which product sold the most, which month had the highest sales, and whether sales are increasing or decreasing.

In education, data analysis can help answer questions such as how many students passed, what the average mark was, which question was answered poorly, and which students may need extra support.

2. Data, information, and insight

Data refers to raw facts. These facts may be numbers, words, dates, categories, or measurements.

Student	Mark
Aisha	72
Ben	45
Carla	81
Daniel	58

Information is created when data is processed or organised. For example: The average mark is 64%.

Insight is created when information is interpreted. For example: Most students passed, but one student may need additional support.

Simple distinction

Data = raw facts. Information = processed data. Insight = an interpretation that helps someone understand or decide.

3. What is a dataset?

A dataset is a structured collection of data. In beginner data analysis, a dataset is often arranged as a table with rows and columns.

StudentID	Name	Programme	Mark	Attendance
1001	Aisha	Diploma IT	72	90
1002	Ben	Diploma IT	45	60
1003	Carla	Diploma IS	81	95
1004	Daniel	Diploma IT	58	75

In this dataset, each row represents one student record. Each column represents one variable or field. Each cell contains one value.

Term	Meaning
Dataset	A collection of data.
Row	One record or observation.
Column	One variable or field.
Value	A single item of data in a cell.
Variable	Something that can change, such as mark, age, date, or category.
Observation	One case, person, item, event, or transaction in the dataset.
Missing value	A value that was not recorded.
Duplicate	A repeated row or repeated value.
Data type	The kind of data stored, such as text, number, or date.

4. Where do datasets come from?

Before we can analyse data, we need a dataset. Datasets can come from many different sources.

Understanding the source is important because it affects how much we can trust the data and what we are allowed to do with it.

Source	Examples and notes
Manually captured data	Student attendance registers, marks entered by lecturers, customer details, stock counts, forms, and surveys. This data often contains typing mistakes, blanks, or inconsistent spelling.
CSV files	Plain text files that store table-like data, often separated by commas. CSV files are simple and

	widely used.
Excel files	Workbooks used by many organisations. They may contain multiple worksheets, formulas, blank rows, or merged cells.
Databases	Systems such as student information systems, online shopping systems, hospital systems, library systems, and payroll systems.
Online systems and websites	Learning management systems, online forms, web traffic, social media activity, and mobile applications.
Sensors and devices	Weather sensors, GPS devices, fitness watches, security systems, manufacturing machines, and medical devices.
Public open datasets	Datasets published by governments, universities, research repositories, and data portals for public use.

Examples of public dataset sources that can be useful for practice include the UCI Machine Learning Repository, Kaggle, Statistics South Africa, government open-data portals, the World Bank, and other national statistical sources. Public datasets must still be checked for reliability, date, licence, meaning of columns, completeness, and privacy concerns.

5. Dataset quality and responsibility

Before analysing a dataset, ask questions about the source, structure, quality, and ethics of the data.

Source questions

- Where did the dataset come from?
- Who created it?
- Why was it created?
- When was it collected?
- Is the source reliable?

Structure questions

- How many rows and columns does it have?
- What does each row represent?
- What does each column represent?
- Are the column names clear?

Quality questions

- Are there missing values?
- Are there duplicate rows?
- Are the values consistent?
- Are the data types correct?
- Are there unusual or impossible values?

Ethics and privacy questions

- Does the dataset contain personal information?
- Do we have permission to use the data?

- Should names or student numbers be removed before sharing?

Responsible analysis

A dataset may contain personal or sensitive information. Data analysis should be done carefully, legally, and ethically.

6. Wide data and long data

The same information can sometimes be arranged in different table layouts. Two common layouts are wide data and long data. The layout affects how easy the data is to filter, group, compare, and visualise.

6.1 Wide data

In wide data, repeated measurements are often stored in separate columns. This layout is often easy for humans to read because it looks similar to a spreadsheet report.

Student	Test1	Test2	Test3
Aisha	65	70	75
Ben	50	55	60
Carla	80	82	85

In this example, each student appears once and each test has its own column. Wide data is often good for reading and reporting, but it can become less convenient when we need to group, filter, compare, or visualise repeated measurements.

6.2 Long data

In long data, repeated measurements are stored in rows instead of separate columns.

Student	Test	Mark
Aisha	Test1	65
Aisha	Test2	70
Aisha	Test3	75
Ben	Test1	50
Ben	Test2	55
Ben	Test3	60
Carla	Test1	80
Carla	Test2	82
Carla	Test3	85

In this example, a student can appear more than once. The test name is stored in one column and the mark is stored in one column. Long data is often useful for analysis because it is easier to group by test, calculate averages per test, and create charts.

Feature	Wide data	Long data
Layout	Repeated values are stored in separate columns.	Repeated values are stored in rows.
Rows and columns	Usually fewer rows and more columns.	Usually more rows and fewer columns.
Good for	Reports and compact spreadsheet-style summaries.	Filtering, grouping, comparing, and visualising.
Beginner rule	Often good for reading.	Often good for analysing.

Rule of thumb

Wide data is often good for reading. Long data is often good for analysing.

7. The basic data analysis workflow

The workflow below shows how raw data is gradually turned into useful findings. In practice, these steps are connected. An analyst may move back to an earlier step when a problem is discovered.

Step 1: Get the data

The data may come from a CSV file, Excel file, database, website, survey, sensor, system export, or public data portal. At this point, the analyst should also ask where the data came from, who created it, when it was collected, and whether permission exists to use it.

Typical activities: download the file, export the data from a system, copy the dataset into the project folder, check the file name, and make sure the file can be opened.

Step 2: Inspect the data

Inspection means looking carefully at the dataset before changing it. The goal is to understand the structure and condition of the data. We check the number of rows and columns, column names, data types, missing values, duplicate rows, unusual values, and whether each column means what we think it means.

Typical Pandas commands for inspection include `df.head()`, `df.shape`, `df.columns`, `df.info()`, `df.describe()`, `df.isna().sum()`, and `df.duplicated().sum()`. In this first lesson, inspection is the most important practical skill.

Important idea: do not start cleaning or analysing before inspecting the dataset. If you do not understand the data, you may clean the wrong thing or calculate an answer that is not trustworthy.

Step 3: Clean the data

Cleaning means fixing or dealing with data quality problems. Common cleaning tasks include removing duplicate rows, handling missing values, correcting spelling inconsistencies, removing unnecessary spaces, fixing incorrect values, and changing columns to the correct data type.

Cleaning should always be justified. For example, if the same student record appears twice, totals and averages may be incorrect. If a mark is missing, we must decide whether to leave it missing, correct it from another source, or remove the record from a specific calculation.

Step 4: Prepare or transform the data

Preparation means shaping the data so that it is easier to analyse. This can include selecting only the columns needed, creating new columns, changing wide data into long data, combining categories, extracting parts of a date, or calculating values such as percentages.

Example: from a Mark column, we could create a PassStatus column. From a Date column, we could create Month or Year. From wide test columns, we could create a long table with Test and Mark columns.

Step 5: Analyse the data

Analysis means using calculations, summaries, filters, comparisons, and grouping to answer the original question. Examples include calculating the average mark, finding the highest and lowest mark, counting how many students passed, or comparing average marks by programme.

A calculation is not automatically an insight. The analyst must still ask whether the result is reasonable and whether data quality problems may have affected it.

Step 6: Visualise the data

Visualisation means using charts or graphs to make patterns easier to see. A table of numbers may be difficult to interpret, but a chart can quickly show differences, trends, or unusual values.

For example, a bar chart can compare average marks by programme, while a histogram can show how marks are distributed. Visualisation will be introduced in more detail in later lessons.

Step 7: Communicate the findings

Communication means explaining the result clearly and responsibly. A correct calculation is not useful unless the analyst can explain what was found, what it means, and what limitations should be considered.

Good communication usually includes the question asked, the data used, the method followed, the result, the interpretation, and any important warnings such as missing values or possible bias.

Step 8: Review and document the work

Review means checking whether the result makes sense and whether the steps were correct.

Documentation means keeping notes about what was done and why. This helps another person understand or repeat the analysis later.

Beginner reminder: In this first lesson, we focus mainly on getting the data and inspecting it. Cleaning, transformation, grouping, and visualisation will be developed in later lessons.

Workflow summary

Get the data -> Inspect the data -> Clean the data -> Analyse the data -> Visualise the data -> Communicate the findings

8. Why use Python and Pandas?

Python is a popular programming language used for software development, automation, data analysis, artificial intelligence, and web development. For data analysis, Python is useful because it has powerful libraries for working with data and can automate repeated analysis tasks.

Pandas is a Python library used for working with structured data. Structured data is data arranged in rows and columns, similar to a spreadsheet or database table. The main Pandas object for beginners is the DataFrame, which is a table-like structure with rows and columns.

Pandas can load data from files, inspect datasets, select rows and columns, clean data, calculate summaries, group data, reshape data, and prepare data for charts and reports.

9. First Pandas example: loading and inspecting a dataset

To use Pandas, we first import it. The abbreviation `pd` is commonly used by Python programmers when working with Pandas.

```
import pandas as pd
```

Suppose we have a CSV file called `students.csv`. We can load it like this:

```
import pandas as pd

df = pd.read_csv("students.csv")
```

In this example, `pd` refers to the Pandas library, `read_csv()` reads the CSV file, `students.csv` is the file name, and `df` is the variable that stores the dataset as a DataFrame.

10. Basic inspection commands

Command	Purpose
<code>df.head()</code>	Displays the first five rows by default.
<code>df.shape</code>	Shows the number of rows and columns.
<code>df.columns</code>	Shows the column names.
<code>df.info()</code>	Shows column names, non-missing values, data types, and memory usage.
<code>df.describe()</code>	Shows summary statistics for numeric columns.
<code>df["Mark"].mean()</code>	Calculates the average value in the Mark column.
<code>df["Mark"].max()</code>	Finds the highest value in the Mark column.
<code>df["Mark"].min()</code>	Finds the lowest value in the Mark column.

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df.head())
print(df.shape)
print(df.columns)
print(df.info())
print(df.describe())

average_mark = df["Mark"].mean()
highest_mark = df["Mark"].max()
lowest_mark = df["Mark"].min()

print("Average mark:", average_mark)
print("Highest mark:", highest_mark)
print("Lowest mark:", lowest_mark)
```

Interpretation

If the average mark is 64.2, do not only report the number. Explain it: The average student mark in this dataset is 64.2%.

11. Practical activity

Download the file `students.csv` and save it in the same folder as your Python file. Then complete the following tasks.

1. Import Pandas.
2. Load the file students.csv.
3. Display the first five rows.
4. Display the shape of the dataset.
5. Display the column names.
6. Display information about the dataset.
7. Display summary statistics.
8. Calculate the average mark.
9. Identify the highest mark.
10. Identify the lowest mark.
11. Write one sentence explaining what you learned from the dataset.

Workflow reflection

12. Which workflow phase did you practise when you used `df.head()`, `df.shape`, `df.columns`, `df.info()`, and `df.describe()`?
13. Why should inspection happen before cleaning?
14. Name one possible cleaning problem that might appear in a student marks dataset.
15. Name one new column that could be created during the preparation or transformation phase.
16. Write one sentence explaining why communication is part of data analysis.

12. Wide vs long activity

Open `student_tests_wide.csv` and `student_tests_long.csv`. Compare the two files. They contain the same type of information, but the layout is different.

17. Which file is in wide format?
18. Which file is in long format?
19. In the wide file, where are the test names stored?
20. In the long file, which column stores the test name?
21. Which layout would be easier if you wanted to calculate the average mark for each test?
22. Explain in your own words why the shape of a dataset matters.

13. Self-check questions

23. What is data analysis?
24. What is the difference between data, information, and insight?
25. What is a dataset?
26. What does one row usually represent?
27. What does one column usually represent?
28. Name three possible dataset sources.
29. What is a CSV file?
30. Why should we ask where a dataset came from?
31. Why is privacy important when working with datasets?
32. What is wide data?
33. What is long data?
34. Which layout is often easier for grouping and visualisation?
35. What is Pandas?

36. What is a DataFrame?
37. What does `df.head()` do?
38. What does `df.shape` show?
39. Why is interpretation important in data analysis?
40. Which workflow phase comes before cleaning?
41. Why is data inspection important?
42. What is one example of a data cleaning task?
43. What is the difference between analysing data and communicating findings?
44. Why may an analyst need to return to an earlier workflow phase?

14. Lesson summary

- Data analysis is the process of using data to answer questions and support decisions.
- Datasets are often arranged in rows and columns.
- Datasets can come from manual capture, CSV files, Excel files, databases, online systems, sensors, and public data portals.
- Dataset quality, privacy, and source reliability matter.
- Wide data stores repeated measurements in separate columns.
- Long data stores repeated measurements in rows and is often easier for analysis.
- Python and Pandas help us load, inspect, clean, analyse, and present data.
- The first Pandas skill is loading a dataset and inspecting it carefully.
- The data analysis workflow is connected and sometimes iterative, not always a straight line.
- Inspection helps us understand the structure and quality of the data before cleaning or analysis.
- Cleaning and preparation improve the usefulness and reliability of the dataset.
- Analysis produces results, while interpretation and communication explain what those results mean.

References and useful dataset sources

- pandas project: <https://pandas.pydata.org/>
- pandas IO tools documentation: https://pandas.pydata.org/docs/user_guide/io.html
- pandas reshaping documentation: https://pandas.pydata.org/docs/user_guide/reshaping.html
- UCI Machine Learning Repository: <https://archive.ics.uci.edu/>
- Statistics South Africa: <https://www.statssa.gov.za/>
- Stats SA ISIBalo Data Portal: <https://isibaloweb.statssa.gov.za/>
- Kaggle Datasets: <https://www.kaggle.com/datasets>
- World Bank Open Data: <https://data.worldbank.org/>

13. Workflow checkpoint activity

Use the workflow to decide where each action belongs. This activity matches the workflow checkpoint slide and helps prepare students for the practical exercise.

Place each action into the correct workflow phase:

- Load `students.csv` into a DataFrame.
- Run `df.info()` to inspect data types and missing values.
- Remove a duplicate row.
- Create a new column called `PassStatus`.

- Calculate the average mark.
- Write one short conclusion explaining what the result means.

Discussion questions

- Which action belongs to the inspection phase?
- Why should inspection happen before cleaning?
- Which actions belong to preparation, analysis, and communication?
- Can an analyst return to an earlier step? Explain your answer.

14. Keep the materials connected

The slides, practical exercise, and handout follow the same sequence: dataset sources, data quality, workflow, wide versus long data, and first Pandas inspection commands. Use this shared sequence to revise the lesson.